

Eurostat data analysis

Geospatial data management project

Federico Garegnani

Università degli Studi di Milano

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Presentation of the datasets used (1)

3 datasets:

① *Census grid 2021*

- Grid of 1 km^2 cells
- Columns: sex, age group, activity status, place of birth, population, population one year prior the census

② *Geographic accessibility*

- Grid of cells of 1 km^2
- Accessibility as travel time from each cell to the nearest healthcare service
- To catch each grid cell to the road network, the nearest network node from the centroid inside the cell was used
- Time measured as travel by car using the road network, data from TomTom Multinet datasets. Dijkstra algorithm used to calculate the shortest distance



Presentation of the datasets used (2)

3 *Communes*

- Administrative boundaries of municipalities
- Columns: commune id, country id, country code, commune name, ASCII name, true flag, NSI code, NSI name, Latin name, NUTS code

All the datasets use ETRS89 as reference system.



Filtering the datasets

Definition of table `it_communes` and indexes for all the tables.

```
CREATE TABLE it_communes AS (  
    SELECT *  
    FROM "all_communes" c  
    WHERE "CNTR_CODE" = 'IT'  
);  
  
CREATE INDEX it_communes_idx  
ON it_communes USING GIST (geom);  
...
```



Calculation of the population by municipality

The formula used:

$$P_C = P_G \sum \frac{A(C \cap G)}{A(G)} \quad (1)$$

and the query performed:

```
WITH pop_calc AS (  
  SELECT  
    c."COMM_ID",  
    ROUND(SUM(ST_Area(ST_Intersection(g.geom, c.geom)) / 1000000  
      ↪ * g."T")) AS pop  
  FROM census_2021 g JOIN it_communes c ON ST_Intersects(g.geom,  
    ↪ c.geom)  
  GROUP BY c."COMM_ID"  
)  
UPDATE it_communes c  
SET pop = p.pop  
FROM pop_calc p  
WHERE c."COMM_ID" = p."COMM_ID";
```



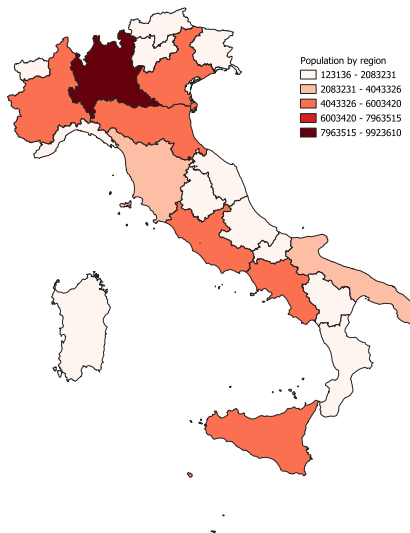
Calculation of mean accessibility value

Computation of the mean accessibility times by commune to the closest healthcare structure and to the 3 closest ones. Years 2020 and 2023.

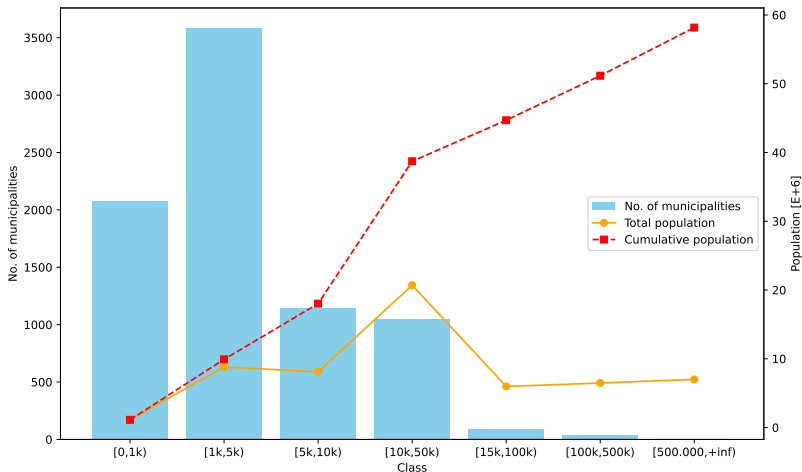
```
WITH avg_t AS (  
  SELECT  
    c."COMM_ID",  
    AVG(g.health_2020_n1) AS mean_health_2020_n1,  
    AVG(g.health_2020_n3) AS mean_health_2020_n3,  
    ...  
  FROM it_communes c JOIN grid_accessibility_health g ON  
    ↪ st_intersects(c.geom, g.geom)  
  GROUP BY c."COMM_ID"  
)  
UPDATE it_communes as c  
SET  
  mean_health_2020_n1 = ROUND(t.mean_health_2020_n1::NUMERIC, 2),  
  ...  
FROM avg_t t  
WHERE t."COMM_ID"=c."COMM_ID";
```



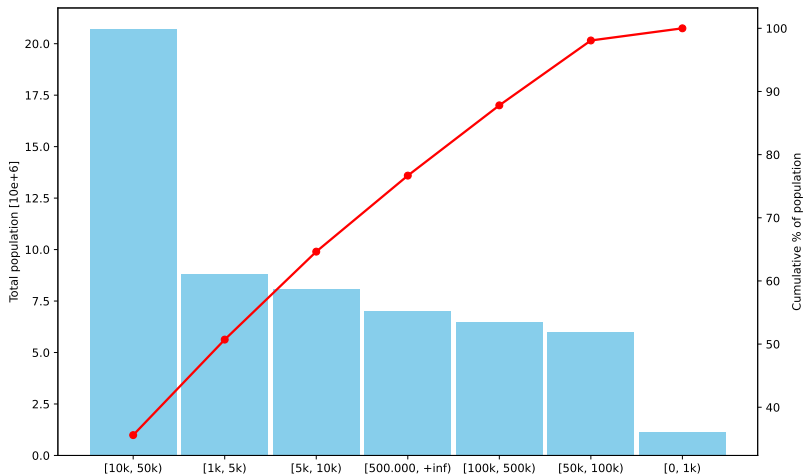
Population by region



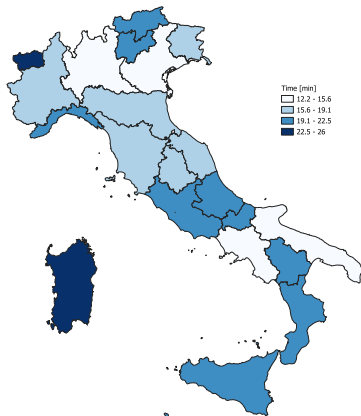
Number of communes by size



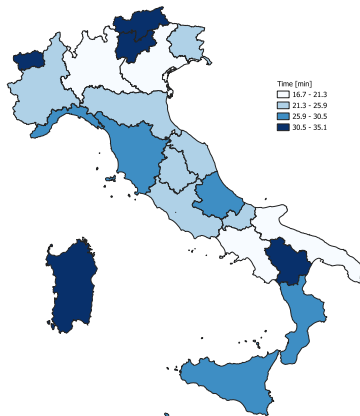
Italian population by size of municipality



Accessibility to healthcare services in Italy



(a) Acc. to the closest centre



(b) Acc. to the 3 closest centres



Categories based on travel time

Access time divided in 4 categories:

No.	Travel time [min]	Accessibility
1	< 15	High
2	$[15, 30)$	Medium-high
3	$[30, 45)$	Medium-low
4	> 45	Low



Accessibility in the North of Italy (1)

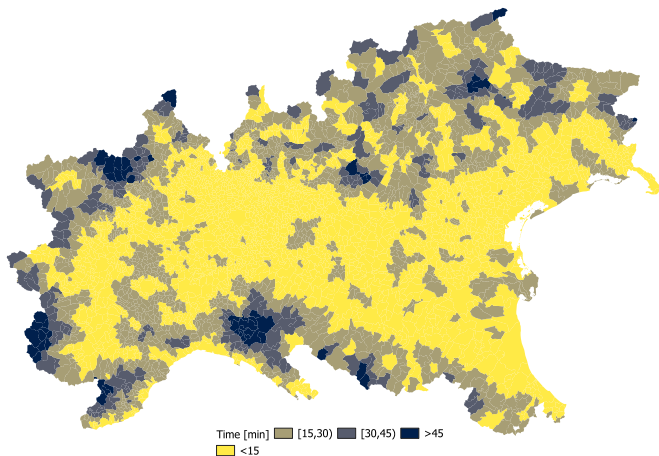


Figure: Accessibility to the closest healthcare service



Accessibility in the North of Italy (2)

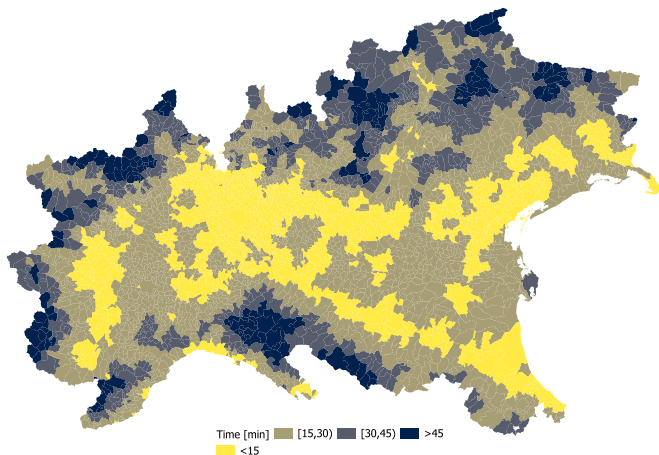


Figure: Accessibility to the 3 closest healthcare services



Distribution of municipalities by class

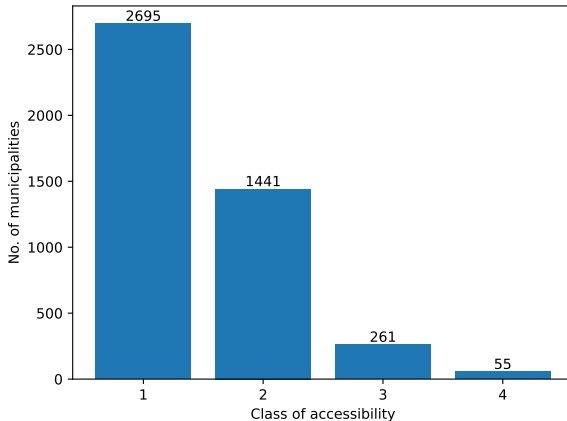


Figure: Time to reach the closest healthcare centre



Population in low accessibility areas

Share of population by accessibility area:

Class	Travel time	Population	% of total
1	< 15	18 836 153	69.32
2	[15, 30)	7 399 725	27.23
3	[30, 45)	792 058	2.91
4	> 45	143 979	0.53
		27 171 915	



Change in healthcare accessibility

7 intervals of change have been defined:

Class	Interval	Communes	
	%	No.	%
Sharp improvement	≤ -20	135	3.03
Relevant improvement	$(-20; -10]$	348	7.82
Slight improvement	$(-10; -5]$	586	13.16
No change	$(-5; 5)$	2298	51.62
Slight worsening	$[5; 10)$	494	11.1
Relevant worsening	$[10; 20)$	355	7.97
Sharp worsening	≥ 20	236	5.3
		4452	



Variation between 2020 and 2023

Computed with this formula:

$$\Delta_{n1}[\%] = \frac{\bar{t}_{2023n1} - \bar{t}_{2020n1}}{\bar{t}_{2020n1}} * 100$$

where

Δ Variation as percentage

$\bar{t}$ Mean access time in the specified year

Thus:

$\Delta > 0 \Rightarrow$ worsening

$\Delta < 0 \Rightarrow$ improvement



Formula implementation in SQL

The query executed to compute accessibility variation between 2020 and 2023:

```
SELECT
  c."COMM_ID",
  c."COMM_NAME",
  c.mean_health_2020_n1,
  c.mean_health_2023_n1,
  ROUND(((c.mean_health_2023_n1 - c.mean_health_2020_n1) /
    ↪ c.mean_health_2020_n1 * 100.0)::NUMERIC, 2) AS
    ↪ variation_n1,
  c.geom
FROM it_communes c
WHERE SUBSTRING(c."NUTS_CODE" FROM 1 FOR 3) IN ('ITC', 'ITH');
```



Map of changes

